Parameter Calibration for Reduced-Bandwidth Two-Photon Waveguide-QED Simulation

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Quantum Optics • Optical Quantum Computing • Quantum Networking • Numerical Simulation • Waveguide QED

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Outline

1 Introduction and Motivations

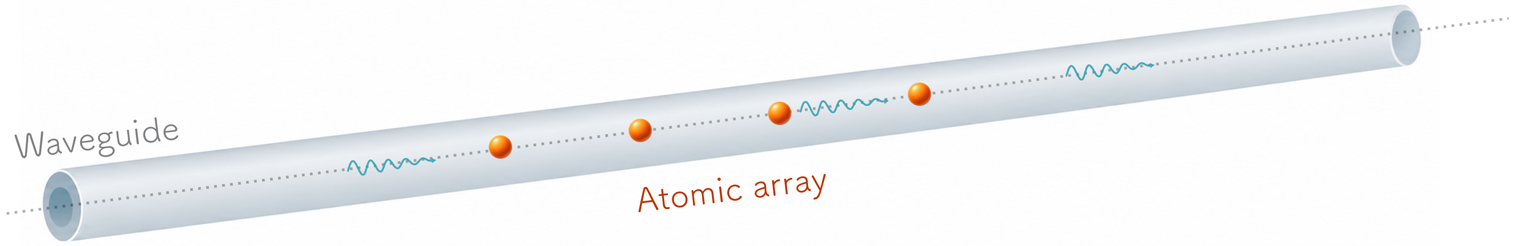
2 Methodology

3 Results

4 Conclusion and Future Perspectives

Introduction and Motivations

Waveguide QED platforms in quantum computing and networking



Matter system (e.g. atom, superconducting circuit)

Photon

Quantum communication networks[1,2,3]

- Interfaces for **processing traveling photons** at quantum nodes
- Nanofiber interfaces for **high-rate remote entanglement generation**

Quantum computation[4]

- Going beyond linear-optics photonic quantum computing, exploiting **effective photon-photon interaction**

Quantum simulation[1,2]

- **Photonic simulators** of quantum many-body dynamics

1. A. S. Sheremet et al., “Waveguide quantum electrodynamics: Collective radiance and photon-photon correlations,” Oct. 2022
2. D. Roy et al., “Strongly interacting photons in one-dimensional continuum,” May 2017.
3. S. Sunami et al., “Scalable Networking of Neutral-Atom Qubits: Nanofiber-Based Approach for Multiprocessor Fault-Tolerant Quantum Computers,” Feb. 2025.
4. H. Zheng et al., “Waveguide-QED-Based Photonic Quantum Computation”, Aug. 2013

Introduction and Motivations

Hamiltonian model for traveling photons

- The dynamics of the system is described with an energy function, the **Hamiltonian**
  - **Travelling photons:** collection of frequency modes
  - **Matter:** two-level system (TLS) with a single transition frequency
  - **Interaction:** excitation exchange between matter and photons
- The model has two hyperparameters[1]: the TLS **transition frequency** and **decay rate**

Hamiltonian modeling

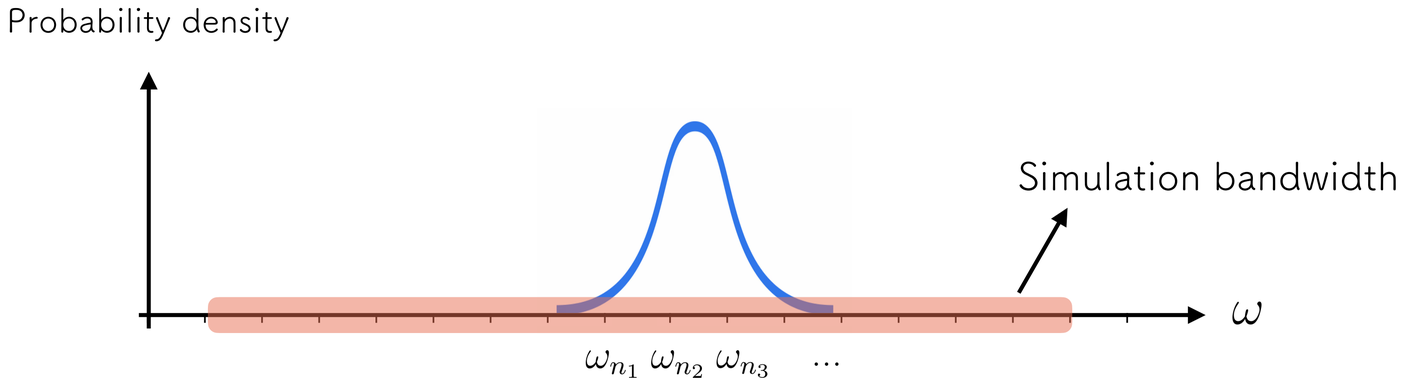
$$H\Big[\omega_0, \gamma_0\Big] = \sum_{\alpha\in\{L,R\}} \int \frac{d\omega}{d\pi} \omega\, a^\dagger_\omega a_\omega + \omega_0 |e\rangle\langle e| + i\frac{\gamma_0}{2} \sum_{\alpha\in\{L,R\}} \int \frac{d\omega}{d\pi} \Big[a_\omega |e\rangle\langle g| - \text{h.c}\Big]$$

1. D. Roy et al., “Strongly interacting photons in one-dimensional continuum,” May 2017.

Introduction and Motivations

Spectral bandwidth and computational cost

- Numerical simulations retain only a **finite bandwidth** of photonic modes



- **System dimensionality** strongly depends on the **simulated spectral bandwidth**
  - Fock state basis: $|(\alpha_1, \omega_1), (\alpha_2, \omega_2), \sigma\rangle$
  - ▲ **Numerical cost of** $\mathcal{O}\left(|B|^N\right)$

Propagation channels: $\alpha \in \{L, R\}$

TLS state: $\sigma \in \{g, e\}$

Photon frequencies: $\omega \in B$

Introduction and Motivations

Bandwidth reduction in frequency domain

- In this work, we investigate **frequency-domain simulation** of waveguide-QED
  - The spectral content of the simulation is explicit
  - Supports arbitrary light–matter spectral densities
  - Here, we begin with a simple, analytically controlled Markovian testbed with flat spectral density
- Practical applications typically operate within **restricted spectral windows**
  - **Optical fibers:** telecom wavelengths around **1550 nm** minimize long-distance losses
  - **Atomic arrays:** photonic excitations are concentrated around **collective resonances**

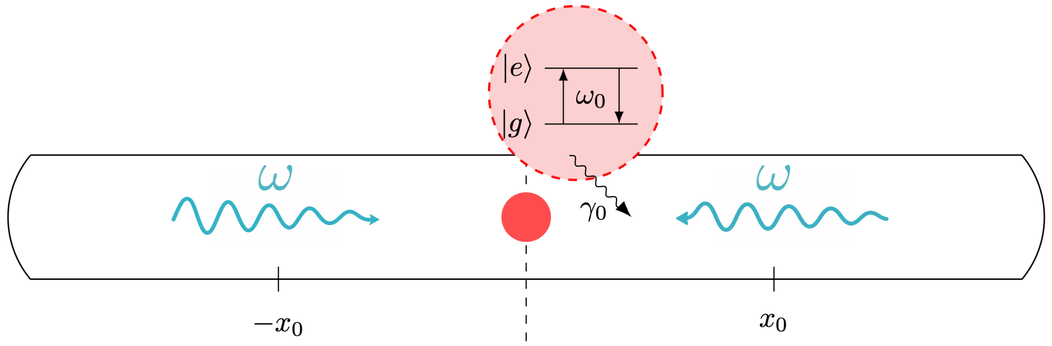
Objective: Simulate only the physically relevant spectral window to **reduce computational cost** while **maintaining accuracy**

2 Methodology

Methodology

Description of the scattering experiment

- Two undistinguishable photons traveling towards a single atom at the center of the waveguide



  - Photons initialized as **monochromatic Gaussian wave-packets**, frequency ω
  - Low atom-photon detuning: $|\omega - \omega_0| \lesssim \gamma_0$
  - Time evolution obtained by **4th-order Runge–Kutta (RK4)** integration of the **Schrödinger equation**

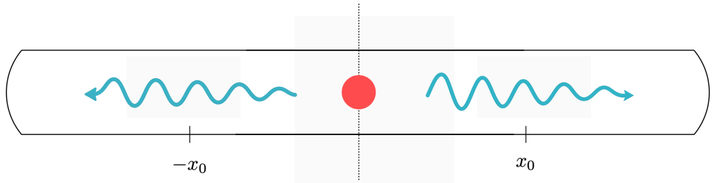
Methodology

Description of the scattering experiment

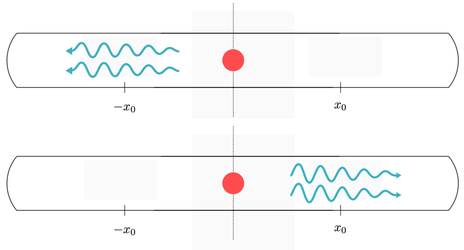
- Extract the probability of photons exiting through different output channels: **coincidence**
  - Known **analytical baseline** in the monochromatic regime

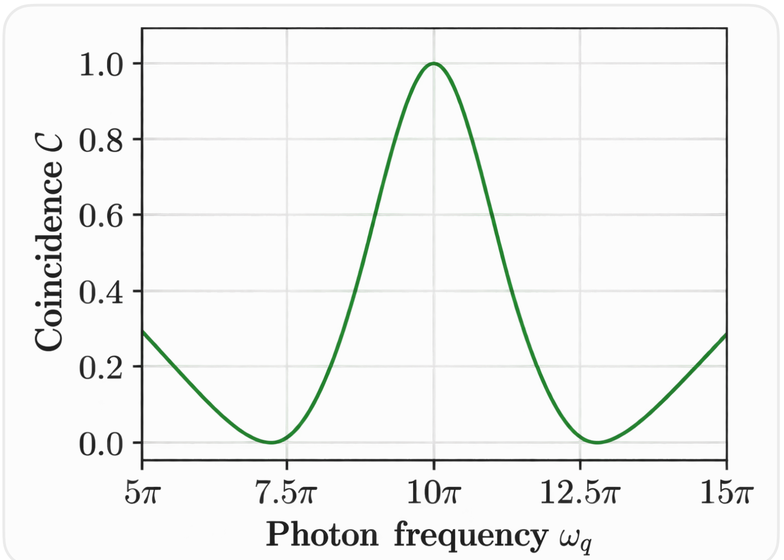
Coincidence in the monochromatic regime

Probability: $\mathcal{C}^{\text{num}}_{\omega_0,\gamma_0}(\omega)$



Probability: $1 - \mathcal{C}^{\text{num}}_{\omega_0,\gamma_0}(\omega)$



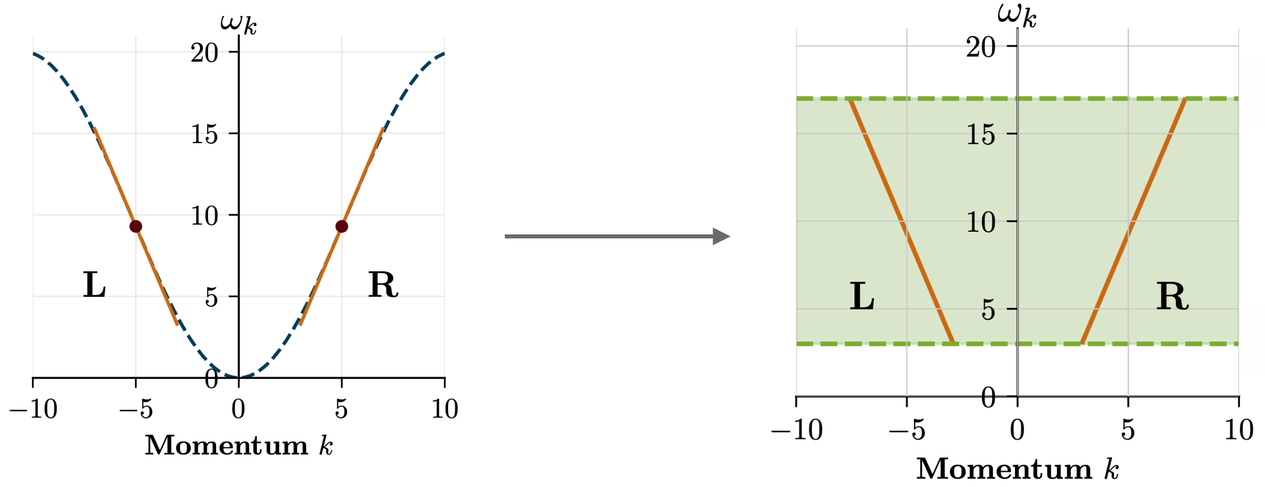


Methodology

Truncation in the frequency domain

- We keep only a **finite portion** of the photon spectrum
  - Retained frequencies: $\omega \in [\omega_{\text{ref}} - \Lambda, \omega_{\text{ref}} + \Lambda]$ Simulation bandwidth
  - Dimensionality scaling: $\mathcal{O}(\Lambda^2)$

Frequency truncation



Methodology

Generic workflow

1 Define the target response

- Fix the physical atomic parameters

$$(\omega_A, \gamma_A)$$

(Experimentally measured in the lab)

- Spectral range of interest

$$[\omega_{\text{ref}} - \Lambda, \omega_{\text{ref}} + \Lambda]$$

(Numerical bandwidth)

2 Parameterize the model

- Adjust the numerical parameters

$$(\omega_0, \gamma_0)$$

Naively, one chooses $(\omega_0, \gamma_0) = (\omega_A, \gamma_A)$

- Define the Hamiltonian

$$H\Big[\gamma_0, \omega_0; (\omega_{\text{ref}}, \Lambda)\Big]$$

3 Run the evolution

- Prepare the initial state
- Fix the time step and the total simulation time
- Generate a sequence of states with RK4

$$|\psi_0\rangle \to |\psi_1\rangle \to \cdots \to |\psi_N\rangle$$
$$t_0 \qquad t_0 + \Delta t \qquad t_0 + N\Delta t$$

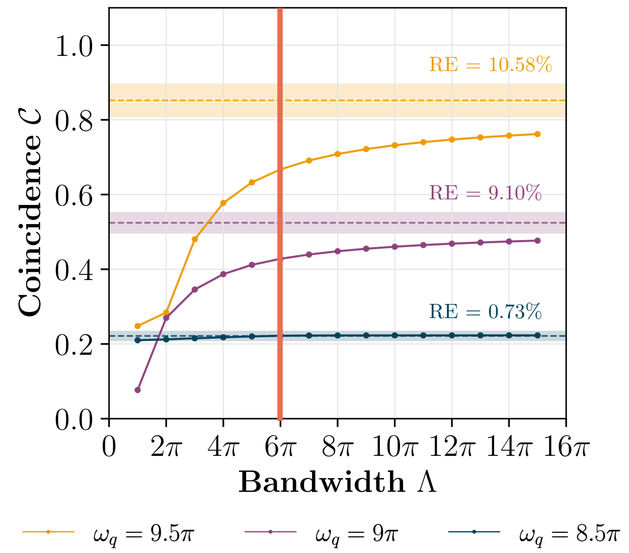
3 Results

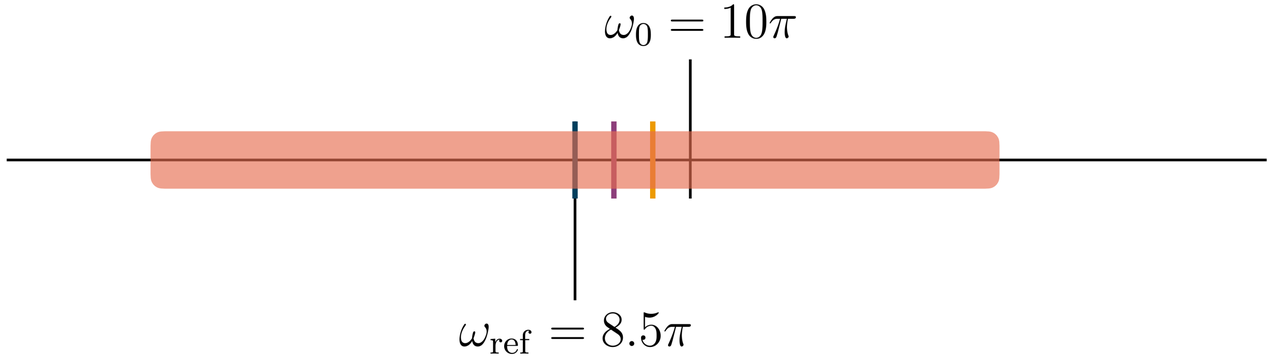
Results

Diagnose truncation effects

- Seek the **minimum bandwidth** required to accurately reproduce the baseline
  - Plot the coincidence against the bandwidth
  - Expectation: convergence towards a plateau (theoretical value)

Coincidence against bandwidth





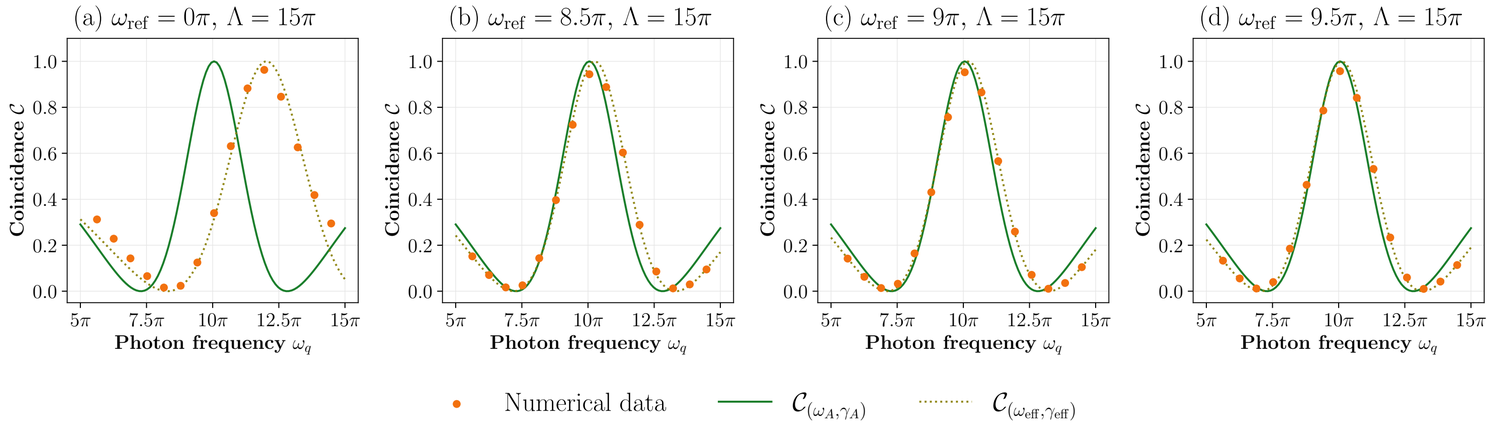
▲ **At such a large bandwidth, all curves should have converged!**

Results

Diagnose truncation effects

- Check accros a broader portion of the numerical spectrum
  - Choose a large bandwidth, and plot the coincidence against the photon frequency

Coincidence against photon frequency



- Dependence on the simulation central frequency: **inconsistent**
- Functional shape preserved: suggests **effective atomic parameters**

Results

Proposing a calibration procedure

- We identify this effect as a **symptom of renormalization**
  - Derivation of **effective parameters** induced by frequency truncation
  - Deduction of a calibration procedure

Effective atomic parameters

$$\lambda = \sum_{k \geq 0} \alpha_k \Big(\omega_0, \gamma_0; \Lambda_{\mathrm{IR}}, \Lambda_{\mathrm{UV}} \Big) \lambda^k$$

Where we defined $\lambda = i(\omega_0 - \omega_A) - \dfrac{\gamma_A}{2}$

Calibration procedure

Require: Physical parameters (ω_A, γ_A)
Ensure: TLS simulator parameters (ω_0, γ_0)
1: Fix ω_{ref}, Λ
2: Compute $\omega_A^{(1)}$ by solving numerically

$$\omega_A = \omega_A^{(1)} - \frac{\gamma_A}{2\pi} \log \left(\frac{\Lambda + (\omega_{\mathrm{ref}} - \omega_A^{(1)})}{\Lambda - (\omega_{\mathrm{ref}} - \omega_A^{(1)})} \right)$$

3: Deduce $\gamma_A^{(1)}$ according to

$$\gamma_A^{(1)} = \gamma_A \left[1 + \frac{\gamma_A}{\pi} \frac{\Lambda}{\Lambda^2 - \left(\omega_A^{(1)} - \omega_{\mathrm{ref}} \right)^2} \right]^{-1}$$

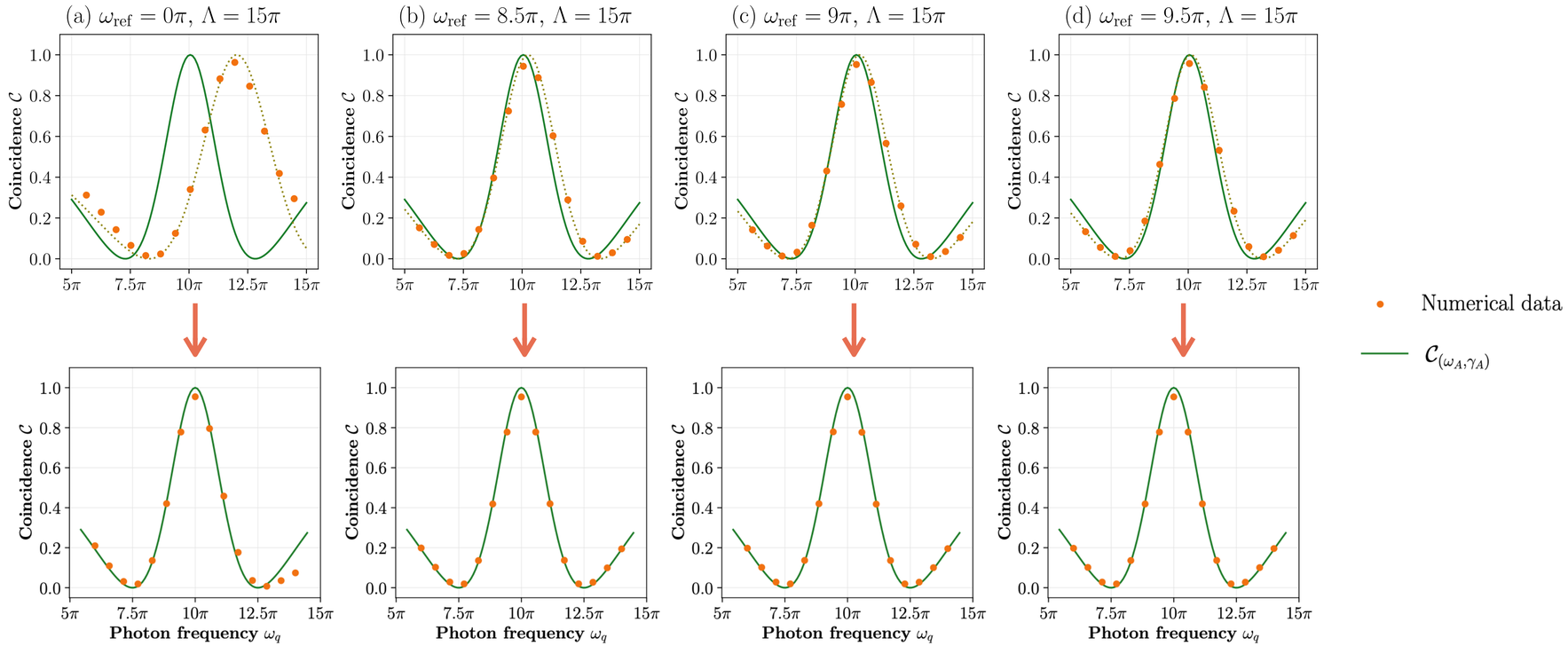
return $(\omega_0, \gamma_0) = (\omega_A^{(1)}, \gamma_A^{(1)})$

Results

Correction of coincidence plots using the calibration procedure

- **Consistency** restored accros coincidence plots: coincidence curve is **accurately reproduced**

Coincidence against photon frequency - **corrected**

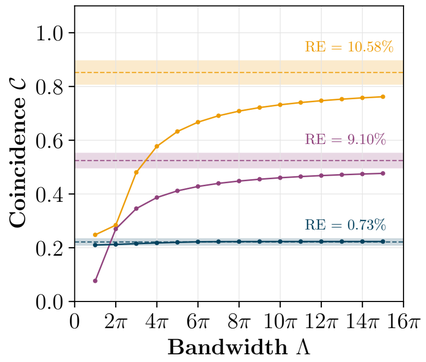


Results

Allowing consistent bandwidth reduction

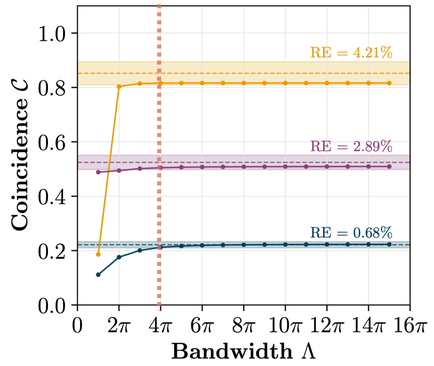
- Calibration enables **consistent bandwidth reduction, lowering computational cost**
  - ▲ Poorly centered frequency windows require larger bandwidths
  - ● Carefully chosen truncation consistently **reduces the number of modes and computational cost**

Coincidence against bandwidth - **corrected**

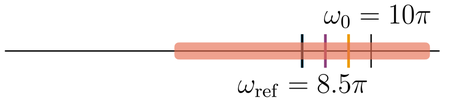


No convergence, even at $\Lambda = 16\pi$

16-fold reduction
of the numerical
Hilbert space



Convergence, even at $\Lambda = 4\pi$

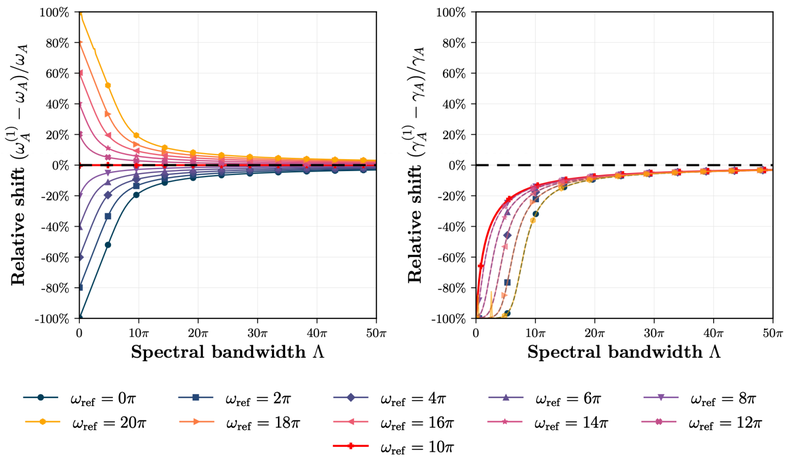


Results

Behavior of the calibration procedure

- Relative correction magnitude **depends on the truncation level**
  - Fix target atomic parameters (ω_A, γ_A)
  - Plot the calibrated parameters $\omega_A^{(1)}(\omega_{\mathrm{ref}}, \Lambda)$ and $\gamma_A^{(1)}(\omega_{\mathrm{ref}}, \Lambda)$ against the bandwidth

Calibrated parameters against bandwidth



- Corrections vanish at large bandwidth but become **significant under strong truncation**
- Larger detuning between the reference and atomic frequencies leads to stronger bandwidth dependence

4 Conclusion and Future perspectives

Conclusion and Future Perspectives

Summary of contributions

- Proposed a **reduced-bandwidth strategy** for waveguide-QED simulations
  - Hard truncation of far-off resonant modes
  - Diagnostic: renormalization effects effectively shift the simulated dynamics
- Demonstrated feasibility on an analytically controlled benchmark using a **calibration procedure**
  - Imposed consistency at the single-photon level: correction of the poles of the projected Green operator in the single photon sector
  - Demonstrated the robustness of the correction on a two-photon scattering experiment

Conclusion and Future Perspectives

Future work

- Extend the calibration procedure to **broader scenarios**
  - Multiple emitters
  - More complex spectral densities
  - Light-matter interactions beyond the waveguide geometry
- Investigate extension to **non-monochromatic photons**

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Thank you for listening